

Linear Algebra Assignment-3

Name : Lana Munir Alam

Roll #: 24K-0573

Class : BCS-3E

Q1) $A = \begin{bmatrix} 4 & 2 & 3 & 3 \\ 0 & 2 & h & 3 \\ 0 & 0 & 4 & 14 \\ 0 & 0 & 0 & 2 \end{bmatrix}$

$\lambda = 4 \rightarrow 2 \text{ dimensional}$

$h = ?$

$(\lambda I - A)x = 0$

substitute value $\lambda = 4$

$(4I - A)x = 0$

$$\left(\begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix} - \begin{bmatrix} 4 & 2 & 3 & 3 \\ 0 & 2 & h & 3 \\ 0 & 0 & 4 & 14 \\ 0 & 0 & 0 & 2 \end{bmatrix} \right) \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

$$\begin{bmatrix} 0 & -2 & -3 & -3 \\ 0 & 2 & -h & -3 \\ 0 & 0 & 0 & -14 \\ 0 & 0 & 0 & 2 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

write it in eq form

$$-2x_2 - 3x_3 - 3x_4 = 0$$

$$2x_2 - hx_3 - 3x_4 = 0$$

$$-14x_4 = 0$$

$$2x_4 = 0$$

$\therefore x_4 = 0$ and x_1 is free variable upto now so $x_1 = s$

substitute x_4 in eq

$$-2x_2 - 3x_3 = 0 \rightarrow \text{eq 1}$$

$$2x_2 - hx_3 = 0 \rightarrow \text{eq 2}$$

$$x_4 = 0$$

$$x_4 = 0$$

writing 2×2 coefficient matrix for (x_1, x_3) :

$$\begin{bmatrix} -2 & -3 \\ \lambda & -h \end{bmatrix}$$

Note:- for eigenspace to have dimension 2. The system must give 2 free variable. We have already gotten x_1 as the first variable. Now we have to find the second variable from among x_2, x_3 . For that we need to get a non-trivial solution (infinitely many solution).

$$\det \begin{bmatrix} -2 & -3 \\ \lambda & -h \end{bmatrix} = 0$$

$$ad - bc = 0$$

$$(-2)(-h) - (\lambda)(-3)$$

$$2h + 6 = 0$$

$$h = -\frac{6}{2}$$

$$h = -3$$

Now for eigenspace for $\lambda = 4$ when $h = -3$

$$\begin{bmatrix} -2 & -3 \\ \lambda & -(-3) \end{bmatrix}$$

RREF

$$\begin{bmatrix} 1 & 3/2 \\ 2 & 3 \end{bmatrix} \quad -\frac{1}{2} R_1$$

$$\begin{bmatrix} 1 & 3/2 \\ 0 & 0 \end{bmatrix} \quad R_2 - 2R_1$$

$$\therefore x_2 = -\frac{3}{2} x_3$$

$$x_3 = t$$

we have found second free variable to be x_3 .

$$\text{So } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ -3/2 \\ 1 \\ 0 \end{bmatrix}$$

Final Answer:

$$h = -3$$

eigen space basis for $\lambda = 4$

$$\{[1, 0, 0, 0]^T, [0, -3/2, 1, 0]^T\}$$

Q2) Finding basis

$$x_1 + x_2 + x_3 + x_4 = 0$$

$$x_1 = -x_2 - x_3 - x_4$$

where x_2, x_3, x_4 are free variables

where

$$x_2 = s$$

$$x_3 = t$$

$$x_4 = u$$

General Solution

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = s \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + u \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$\therefore$ Basis for S is

$$\{[-1, 1, 0, 0]^T, [-1, 0, 1, 0]^T, [-1, 0, 0, 1]^T\}$$

For orthogonal basis:

$$u_1 = [-1, 1, 0, 0]^T$$

$$u_2 = [-1, 0, 1, 0]^T$$

$$u_3 = [-1, 0, 0, 1]^T$$

Basis $\{u_1, u_2, u_3\}$

orthogonal Basis = $\{v_1, v_2, v_3\}$

$$v_1 = u_1$$

$$v_1 = [-1, 1, 0, 0]^T$$

for v_2

$$v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1$$

$$= [-1, 0, 1, 0]^T - \frac{[-1, 0, 1, 0] \cdot [-1, 1, 0, 0]}{(\sqrt{(-1)^2 + (1)^2 + 0^2 + 0^2})^2} \cdot (-1, 1, 0, 0)$$

$$v_2 = [-1, 0, 1, 0]^T - \frac{1}{2} [-1, 1, 0, 0]^T$$

$$v_2 = [-1, 0, 1, 0]^T + \left[\frac{1}{2}, -\frac{1}{2}, 0, 0\right]^T$$

$$v_2 = \left[-\frac{1}{2}, -\frac{1}{2}, 1, 0\right]^T$$

for v_3

$$v_3 = u_3 - \frac{\langle u_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_3, v_2 \rangle}{\|v_2\|^2} v_2$$

$$v_3 = [-1, 0, 0, 1]^T - \frac{[-1, 0, 0, 1]^T \cdot [-1, 1, 0, 0]^T}{(\sqrt{(-1)^2 + (1)^2 + 0^2 + 0^2})^2} \cdot [-1, 1, 0, 0]^T - \frac{[-1, 0, 0, 1]^T \cdot \begin{bmatrix} -1/2 \\ -1/2 \\ 1 \\ 0 \end{bmatrix}}{(\sqrt{(-1/2)^2 + (-1/2)^2 + 1^2 + 0^2})^2} \cdot \begin{bmatrix} -1/2 \\ -1/2 \\ 1 \\ 0 \end{bmatrix}$$

$$v_3 = [-1, 0, 0, 1]^T - \frac{1}{2} [-1, 1, 0, 0]^T - \frac{1/2}{3/2} \left[-\frac{1}{2}, -\frac{1}{2}, 1, 0\right]^T$$

$$v_3 = [-1, 0, 0, 1]^T + \left[\frac{1}{2}, -\frac{1}{2}, 0, 0\right]^T - \frac{1}{3} \left[-\frac{1}{2}, -\frac{1}{2}, 1, 0\right]^T$$

$$v_3 = \left[-\frac{1}{2}, -\frac{1}{2}, 0, 1\right]^T + \left[\frac{1}{6}, \frac{1}{6}, -\frac{1}{3}, 0\right]^T$$

$$v_3 = \left[-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, 1\right]^T$$

$\therefore$ Orthogonal basis

$$v_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} -1/2 \\ -1/2 \\ 1 \\ 0 \end{bmatrix}$$

$$v_3 = \begin{bmatrix} -1/3 \\ -1/3 \\ -1/3 \\ 1 \end{bmatrix}$$

Answer

Q3)

$$A = \begin{bmatrix} 1 & 2 & 5 \\ -1 & 1 & -4 \\ -1 & 4 & -3 \\ 1 & -4 & 7 \\ 1 & 2 & 1 \end{bmatrix}$$

$u_1 \quad u_2 \quad u_3$

$$A = Q \cdot R$$

$5 \times 3 \quad 5 \times 3 \quad 3 \times 3$

col vectors of A

$$u_1 = [1, -1, -1, 1, 1]^T$$

$$u_2 = [2, 1, 4, -4, 2]^T$$

$$u_3 = [5, -4, -3, 7, 1]^T$$

Gram-Schmidt

$$v_1 = u_1$$

$$v_1 = [1, -1, -1, 1, 1]^T$$

for v_2

$$v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} \cdot v_1$$

$$v_2 = [2, 1, 4, -4, 2]^T - \frac{[2, 1, 4, -4, 2]^T \cdot [1, -1, -1, 1, 1]^T}{(\sqrt{1^2 + (-1)^2 + (-1)^2 + 1^2 + 1^2})^2} \cdot [1, -1, -1, 1, 1]^T$$

$$v_2 = [2, 1, 4, -4, 2]^T - \frac{(-5)}{5} \cdot [1, -1, -1, 1, 1]^T$$

$$v_2 = [2, 1, 4, -4, 2]^T + 1[1, -1, -1, 1, 1]^T$$

$$v_2 = [3, 0, 3, -3, 3]^T$$

for v_3

$$v_3 = u_3 - \frac{\langle u_3, v_1 \rangle}{\|v_1\|^2} \cdot v_1 - \frac{\langle u_3, v_2 \rangle}{\|v_2\|^2} v_2$$

solving

$$\begin{aligned} \frac{\langle u_3, v_1 \rangle}{\|v_1\|^2} \cdot v_1 &= \frac{[5, -4, -3, 7, 1]^T \cdot [1, -1, -1, 1, 1]^T}{5} \cdot [1, -1, -1, 1, 1]^T \\ &= \frac{420}{5} [1, -1, -1, 1, 1]^T \\ &= [84, -84, -84, 84, 84]^T \end{aligned}$$

solving

$$\frac{\langle u_3, v_2 \rangle}{\|v_2\|^2} = \frac{[5, -4, -3, 7, 1]^T \cdot [3, 0, 3, -3, 3]^T}{(\sqrt{3^2 + 0^2 + 3^2 + (-3)^2 + 3^2})^2} \cdot [3, 0, 3, -3, 3]^T$$

$$= \frac{-12 \times 1}{36 \times 3} [3, 0, 3, -3, 3]^T = -\frac{1}{3} [3, 0, 3, -3, 3]^T$$

$$= \cancel{[-1, 0, -1, 1, 1]}^T = [-1, 0, -1, 1, -1]^T$$

$$V_3 = U_3 - \frac{\langle U_3, V_1 \rangle}{\|V_1\|^2} \cdot V_1 - \frac{\langle U_3, V_2 \rangle}{\|V_2\|^2} \cdot V_2$$

$$V_3 = [5, -4, -3, 7, 1]^T - [4, -4, -4, 4, 4]^T - [-1, 0, -1, 1, -1]^T$$

$$\cancel{V_3 = [2, 0, 2, 2, -2]^T}$$

$$V_3 = [5, -4, -3, 7, 1]^T + [-4, 4, 4, -4, -4]^T + [1, 0, 1, -1, 1]^T$$

$$V_3 = [2, 0, 2, 2, -2]^T$$

∴ Orthogonal vectors

$$V_1 = [1, -1, -1, 1, 1]^T$$

$$V_2 = [3, 0, 3, -3, 3]^T$$

$$V_3 = [2, 0, 2, 2, -2]^T$$

Now normalising it to get orthonormal vectors

$$q_1 = \frac{V_1}{\|V_1\|} = \frac{[1, -1, -1, 1, 1]^T}{\sqrt{5}} = \left[\frac{1}{\sqrt{5}}, -\frac{1}{\sqrt{5}}, -\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right]^T$$

$$q_2 = \frac{V_2}{\|V_2\|} = \frac{[3, 0, 3, -3, 3]^T}{\sqrt{36}} = \left[\frac{1}{2}, 0, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2} \right]^T$$

$$q_3 = \frac{V_3}{\|V_3\|} = \frac{[2, 0, 2, 2, -2]^T}{\sqrt{16}} = \left[\frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2} \right]^T$$

$$\therefore \text{Matrix } Q = \begin{bmatrix} q_1 & q_2 & q_3 \\ \frac{1}{\sqrt{5}} & \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{\sqrt{5}} & 0 & 0 \\ -\frac{1}{\sqrt{5}} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{\sqrt{5}} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{\sqrt{5}} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

For Matrix $R_{3 \times 3}$ which is an upper triangular matrix

$$R = \begin{bmatrix} \langle u_1, q_1 \rangle & \langle u_2, q_1 \rangle & \langle u_3, q_1 \rangle \\ 0 & \langle u_2, q_2 \rangle & \langle u_3, q_2 \rangle \\ 0 & 0 & \langle u_3, q_3 \rangle \end{bmatrix}$$

So $\langle u_1, q_1 \rangle = [1, -1, -1, 1, 1]^T \cdot [\frac{1}{\sqrt{5}}, -\frac{1}{\sqrt{5}}, -\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}] = \sqrt{5}$

$$\langle u_2, q_1 \rangle = [2, 1, 4, -4, 2]^T \cdot [\frac{1}{\sqrt{5}}, -\frac{1}{\sqrt{5}}, -\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}] = -\sqrt{5}$$

$$\langle u_3, q_1 \rangle = [5, -4, -3, 7, 1]^T \cdot [\frac{1}{\sqrt{5}}, -\frac{1}{\sqrt{5}}, -\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}] = 4\sqrt{5}$$

$$\langle u_2, q_2 \rangle = [2, 1, 4, -4, 2]^T \cdot [\frac{1}{2}, 0, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2}] = 6$$

$$\langle u_3, q_2 \rangle = [5, -4, -3, 7, 1]^T \cdot [\frac{1}{2}, 0, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2}] = -2$$

$$\langle u_3, q_3 \rangle = [5, -4, -3, 7, 1]^T \cdot [\frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}] = 4$$

$$R_{3 \times 3} = \begin{bmatrix} \sqrt{5} & -\sqrt{5} & 4\sqrt{5} \\ 0 & 6 & -2 \\ 0 & 0 & 4 \end{bmatrix}$$

Now Finally Calculate QR decomposition

$$A_{5 \times 3} = Q_{5 \times 3} \cdot R_{3 \times 3}$$

$$\begin{bmatrix} 1 & 2 & 5 \\ -1 & 1 & -4 \\ -1 & 4 & -3 \\ 1 & -4 & 7 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{\sqrt{5}} & 0 & 0 \\ -\frac{1}{\sqrt{5}} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{\sqrt{5}} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{\sqrt{5}} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} \sqrt{5} & -\sqrt{5} & 4\sqrt{5} \\ 0 & 6 & -2 \\ 0 & 0 & 4 \end{bmatrix}$$

$$LHS = RHS$$

hence correctly found Q and R

Q4) Data:

$$\det(A) = 25$$

$$\lambda = 5$$

$$\text{Trace} = 10 \quad (\text{sum of diagonal values})$$

$$\det(A - I\lambda) = (\lambda - 5)^2$$

Sol: Finding missing values

$$A_1 = \begin{bmatrix} 9 & 4 \\ & 1 \end{bmatrix}$$

$$\det(A_1) \Rightarrow ad - bc = 25$$

$$9(1) - 4(x) = 25$$

$$-4x = 25 - 9$$

$$x = -4$$

So missing value of A_1 matrix is -4

$$\therefore A_1 = \begin{bmatrix} 9 & 4 \\ -4 & 1 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 10 & 5 \\ -5 & \end{bmatrix}$$

$$\det(A_2) \Rightarrow ad - bc = 25$$

$$10(x) - 5(-5) = 25$$

$$10x = 25 - 25$$

$$x = \frac{0}{10}$$

$$x = 0$$

So missing value of A_2 matrix is 0

$$\therefore A_2 = \begin{bmatrix} 10 & 5 \\ -5 & 0 \end{bmatrix}$$

Checking Trace

$$A_1 = \begin{bmatrix} 9 & 4 \\ -4 & 1 \end{bmatrix}$$

$$\text{sum of diagonal} = 9 + 1 \\ = 10$$

$$A_2 = \begin{bmatrix} 10 & 5 \\ -5 & 0 \end{bmatrix}$$

$$\text{sum of diagonal} = 10 + 0 \\ = 10$$

$\therefore$ Both matrix trace is equal to 10

Check $\lambda = 5$ is repeated eigenvalue

$$\det(A - I\lambda) = (\lambda - 5)^2$$

for A_1 matrix

$$\begin{bmatrix} 9 & 4 \\ -4 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

~~$$\det \begin{bmatrix} 9 & 4 \\ -4 & 1 \end{bmatrix} = \det \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \quad \begin{bmatrix} 9 & 4 \\ -4 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = (\lambda - 5)^2$$~~

~~$$\det \begin{bmatrix} 4 & 4 \\ -4 & -4 \end{bmatrix}$$~~

$$\det \begin{bmatrix} 9-\lambda & 4 \\ -4 & 1-\lambda \end{bmatrix} = (\lambda - 5)^2$$

~~$$ad - bc$$~~

$$(9-\lambda)(1-\lambda) - 4(-4) = (\lambda - 5)^2$$

$$9 - 9\lambda - \lambda + \lambda^2 + 16 = (\lambda - 5)^2$$

$$\lambda^2 - 10\lambda + 25 = (\lambda - 5)^2$$

$$(\lambda - 5)(\lambda - 5) = (\lambda - 5)^2$$

$$(\lambda - 5)^2 = (\lambda - 5)^2$$

as LHS = RHS hence proved for matrix A_1 $\lambda = 5$ is repeated eigen value.

For matrix A_2

$$\det (A_2 - I\lambda) = (\lambda - 5)^2$$

~~for A_2 matrix~~

$$\begin{bmatrix} 10 & 5 \\ -5 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 10 & 5 \\ -5 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$\det \begin{bmatrix} 10-\lambda & 5 \\ -5 & -\lambda \end{bmatrix} = (\lambda - 5)^2$$

$$(10-\lambda)(-\lambda) - (5)(-5) = (\lambda - 5)^2$$

$$\lambda^2 - 10\lambda + 25 = (\lambda - 5)^2$$

$$(\lambda - 5)(\lambda - 5) = (\lambda - 5)^2$$

$$(\lambda - 5)^2 = (\lambda - 5)^2$$

as LHS = RHS hence proved for matrix A_2 $\lambda = 5$ is repeated eigen value

Find Eigenvector with $Ax = 5x$

solve

$$(A - 5I)x = 0$$

$$A_1 - 5I = \begin{bmatrix} 9-5 & 4 \\ -4 & 1-5 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ -4 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 4 \\ -4 & 4 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

RREF

$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ -4 & -4 & 0 \end{array} \right] \xrightarrow{\frac{1}{4}R_1}$$

$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right] R_2 + 4R_1$$

$x_1 \quad x_2$

so $x_1 + x_2 = 0 \rightarrow x_1 = -x_2$
 $x_2 = t$ free var

general solution

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

hence eigenvector for matrix $A_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

for A_2

$$(A_2 - 5I) = \begin{bmatrix} 10-5 & 5 \\ -5 & 0-5 \end{bmatrix} = \begin{bmatrix} 5 & 5 \\ -5 & -5 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 5 \\ -5 & -5 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

RREF

$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ -5 & -5 & 0 \end{array} \right] \xrightarrow{\frac{1}{5}R_1}$$

$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right] R_2 + 5R_1$$

$x_1 \quad x_2$

so

$$\begin{aligned} x_1 + x_2 &= 0 \\ x_1 &= -x_2 \end{aligned} \quad \left| \quad \begin{aligned} x_2 &= t \\ \text{free var} \end{aligned} \right.$$

general solution

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

hence eigenvector for matrix $A_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

The matrix A_1, A_2 are not diagonalizable cause their algebraic multiplicity of $\lambda = 5$ is 2 (since it's repeated). But the

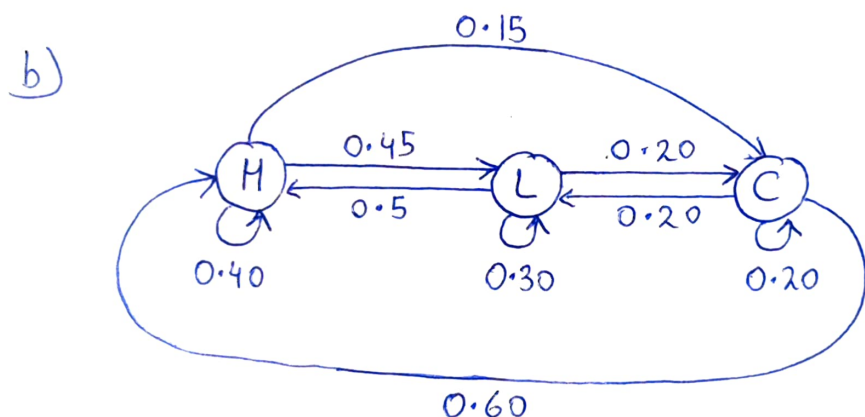
geometric multiplicity is 1 (only one eigen vector per matrix),
 a matrix is diagonalizable only if the geometric multiplicity (GM)
 equals the algebraic multiplicity (AM) for each eigen value.

And as in our case

$GM < AM \therefore$ not diagonalizable.

Q5a) Transition Matrix

$$A = \begin{matrix} & \begin{matrix} H & L & C \end{matrix} \\ \begin{matrix} H \\ L \\ C \end{matrix} & \begin{bmatrix} 0.40 & 0.45 & 0.15 \\ 0.50 & 0.30 & 0.20 \\ 0.60 & 0.20 & 0.20 \end{bmatrix} \end{matrix}$$



c) $P(H \rightarrow H) = 0.40$ probability of using Hacker Rank tomorrow if it was used today.

d) $P^4(C \rightarrow C) = ?$

Initial $x(0) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

Day 1

$$x(1) = P x(0) = \begin{bmatrix} 0.60 \\ 0.20 \\ 0.20 \end{bmatrix}$$

H: 0.60, L: 0.20, C: 0.20

Day 2

$$x(2) = P x(1)$$

$$x(2) = \begin{bmatrix} 0.4 & 0.45 & 0.15 \\ 0.5 & 0.30 & 0.20 \\ 0.60 & 0.20 & 0.20 \end{bmatrix} \cdot \begin{bmatrix} 0.60 \\ 0.20 \\ 0.20 \end{bmatrix}$$

=

d) $P^4(C \rightarrow C) = ?$

$$P^2 = \begin{bmatrix} 0.40 & 0.45 & 0.15 \\ 0.5 & 0.30 & 0.20 \\ 0.60 & 0.20 & 0.20 \end{bmatrix} \cdot \begin{bmatrix} 0.40 & 0.45 & 0.15 \\ 0.5 & 0.30 & 0.20 \\ 0.60 & 0.20 & 0.20 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0.475 & 0.345 & 0.18 \\ 0.47 & 0.355 & 0.175 \\ 0.46 & 0.37 & 0.17 \end{bmatrix}$$

$$P^4 = (P^2)^2 = \begin{bmatrix} 0.4705 & 0.3529 & 0.1764 \\ 0.4706 & 0.3529 & 0.1764 \\ 0.4706 & 0.3529 & 0.1764 \end{bmatrix}$$

so $P^4(C \rightarrow C) = 0.1764$ probability

e)

$$H = 0.4H + 0.5L + 0.6C$$

$$H - 0.4H - 0.5L - 0.6C = 0$$

$$0.6H - 0.5L - 0.6C = 0 \quad \text{eq A}$$

$$H_2 = 0.45H + 0.3L + 0.2C$$

$$H_2 - 0.45H - 0.3L - 0.2C = 0$$

$$-0.45H - 0.7L - 0.2C = 0 \quad \text{eq B}$$

$$H = \frac{0.5L + 0.6C}{0.6} = \frac{5}{6}L + C$$

From B

$$-0.45\left(\frac{5}{6}L + C\right) - 0.7L - 0.2C = 0$$

$$0.325L - 0.65C = 0$$

$$L = \frac{0.6}{0.325} \quad C = 2C$$

Putting L in H

$$\frac{5}{6}(2C) + C = \frac{8}{3}C$$

e)

$$H = 0.4H + 0.5L + 0.6C$$

$$L = 0.45H + 0.3L + 0.2C$$

$$C = 0.15H + 0.2L + 0.2C$$

$$H + L + C = 1$$

} eq 1

$$0.6H - 0.5L - 0.6C = 0$$

} eq 1

$$-0.45H + 0.7L - 0.2C = 0$$

} eq 2

$$-0.15H - 0.2L + 0.8C = 0$$

} eq 3

from 1

$$0.6H = 0.5L + 0.6C$$

$$H = \frac{0.5}{0.6} L + C$$

$$H = \frac{5}{6} L + C$$

substitute H in eq 2

$$-0.45\left(\frac{5}{6}L + C\right) + 0.7L - 0.2C = 0$$

$$-0.375L - 0.45C + 0.7L - 0.2C = 0$$

$$(0.7 - 0.375)L + (-0.45 - 0.2)C = 0$$

$$0.325L - 0.65C = 0$$

$$L = \frac{0.65C}{0.325}$$

$$L = 2C$$

substitute H and L in eq 1

$$H = \frac{5}{6} L + C$$

$$H = \frac{5}{6} (2C) + C$$

$$H = \frac{8}{3} C$$

Using eq 4

$$H + L + C = 1$$

$$\frac{8}{3} C + 2C + C = 1$$

$$C = \frac{3}{17}$$

for L

$$L = 2C$$

$$L = 2\left(\frac{3}{17}\right)$$

$$L = \frac{6}{17}$$

for H

$$H = \frac{8}{3} C$$

$$H = \frac{8}{3} \left(\frac{3}{17}\right)$$

$$H = \frac{8}{17}$$

∴ (H, L, C)

$$\left(\frac{8}{17}, \frac{6}{17}, \frac{3}{17}\right)$$

$$(0.471, 0.353, 0.176)$$